


PHYSICS

PAPER 1

3 HOURS



STUDY SPHERE EXAMINATION BOARD

"Your Partners in Academic Excellence"

Uganda Advanced Certificate of Education

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INSTRUCTIONS TO CANDIDATES:

1. This paper has three sections A,B, and C.
2. Attempt **four** questions in total. Each question carries **20 marks**.
3. You must attempt **one question from each of the three sections**:
4. Section A is compulsory.
5. Use the provided data and support materials to guide your solutions.
6. Clearly show all calculations and assumptions.

SECTION A

Item 1

The display screens of digital devices use Light Emitting Diodes (LEDs), which emit high amounts of blue light. Scientific research has shown that long exposure to such light is harmful to the human eyes. An ICT student of senior five in a certain school is worried that their eyes may be at a risk due to continuous use of computers.

Support Material Provided:

- Wavelength of blue light = 450 nm
- Planck's constant, $h = 6.63 \times 10^{-34}$ Js,
- speed of light in vacuum, $c = 3.0 \times 10^8$ m/s
- Damage threshold for the human eye = 3.1 eV

Tasks:

- Use the information provided to dispel fears of the student.
- Explain to the student some of the symptoms of eye defects which can arise from prolonged exposure to radiations of energy beyond certain threshold.
- Based on principles of Physics, suggest possible protective strategies which the student can employ to safeguard his/her eyes.

SECTION B

Item 2

Many rural areas experience frequent motorcycle accidents due to sharp curves and poor road design. Local council authorities seek physics-based input on safe road redesign.

Support Material Provided:



Radius of curved road = 12 m, Banking angle = 5° , Coefficients of friction: 0.6 (dry), 0.35 (wet).
Mass of motorcycle + rider = 180 kg.

Tasks:

- (a) With justification, advise the local council on the maximum safe speed that the riders should move at, on such roads under wet and dry conditions.
- (b) Suppose that the curved road was not banked, provide convincing advice to the local council authorities, the design that would enable the motorcyclists reach their destinations earlier in both scenarios.

Item 3

A 500 kg satellite in an elliptical orbit (apogee: 42,000 km, perigee: 200 km) requires stabilization. Engineers must calculate orbital parameters to fire thrusters at apogee for circularization. Earth's mass = 5.97×10^{24} kg, radius = 6.4×10^6 m.

Tasks:

- a) State Kepler's third law and derive the orbital period T in terms of G , M , and semi-major axis r .
- b) Calculate the satellite's angular momentum at perigee (velocity = 8.2 km/s).
- c) Show that the total mechanical energy, $E = -\frac{GMm}{2r}$ and find the energy input needed to circularize the orbit.

SECTION C**Item 4**

A new rural health center is built in a valley where mobile phone signals are weak. You are part of a student team challenged to design a light-based communication system between the health center and a village watch tower located 150 meters away on a hill, in order to ease communication at the health centre. Hint: refractive indices of glass and air are 1.5 and 1.0, respectively.

Tasks:

- a) Explain how communication occurs from the watch tower to the health center through the medium identified.

Item 5

Market vendors have over the years complained of destructive fire outbreaks caused by overloaded electric power extension cables. These fire outbreaks have often left market vendors devastated due to loss of property. To alleviate this situation, the chairman of the market vendors' association requested you, an A'level student of physics, to evaluate and recommend safer electrical practices.

During your evaluation, you find out that: the market is wired using an electrical cable of length 20m of resistance $0.5 \Omega \text{ m}^{-1}$ carrying a current of 8A for a duration of 4 hours per day. Maximum safe energy dissipation by the cable is 331.2 kJ m^{-1} .

Tasks:

- (a) Analyze the information provided and suggest the cause of fire in the market.
- (b) Propose practical safety improvements and awareness strategies to the vendors to avoid such occurrence.

***** THE END *****